

Q1

To adjust the emulated CPU speed, use **ctrl-F11** and **ctrl-F12**.
To activate the keymapper **ctrl-F1**.
For more information read the **README** file in the DOSBox directory.

HAVE FUN!
The DOSBox Team <http://www.dosbox.com>

```
Z:\>SET BLASTER=A220 I7 D1 H5 T6

Z:\>MOUNT C "."
Drive C is mounted as local directory ./

Z:\>C:

C:\>CALL EXECUTE.BAT

C:\>nasm code.asm -o proj.com

C:\>proj.com

msg1: Hi! I am Saaif Suleman.
msg2: I am from Lahore.
msg3: I Study at FAST.
msg4: My Roll No is 24L-2606#.
C:\>
```

```
[org 0x0100]
jmp start
msg1 db 'msg1: Hi! I am Saaif Suleman.$'
msg2 db 'msg2: I am from Lahore.$'
msg3 db 'msg3: I Study at FAST.$'
msg4 db 'msg4: My Roll No is 24L-2606#.$'
```

```
WaitKey:
    mov ah, 1
    int 21h
    ret
```

```
PrintMsg:
    mov ah, 9
    int 21h
    ret
```

```
start:
```

```
call WaitKey  
mov dx, msg1  
call PrintMsg
```

```
call WaitKey  
mov dx, msg2  
call PrintMsg
```

```
call WaitKey  
mov dx, msg3  
call PrintMsg
```

```
call WaitKey  
mov dx, msg4  
call PrintMsg
```

```
mov ax, 0x4c00  
int 21
```

Q2

[org 0x0100]

jmp code

w equ 50 ; width offset

x equ 50 ; starting x coordinate of line

y equ 100 ; starting y coordinate of line

c equ 60 ; color

x1 equ 50 ; starting x coordinate of line

y1 equ 100 ; starting y coordinate of line

code:

mov ah, 0

mov al, 13h

int 10h

; draw diagonal 11:

mov cx, x

mov dx, y

mov al, c

u1:

inc dx

mov ah, 0ch ; put pixel

int 10h

inc cx

cmp cx, x+w

jbe u1

mov cx, x

mov dx, y+1

mov al, c

u2:

dec dx

mov ah, 0ch ; put pixel

int 10h

inc cx

cmp cx, x+w

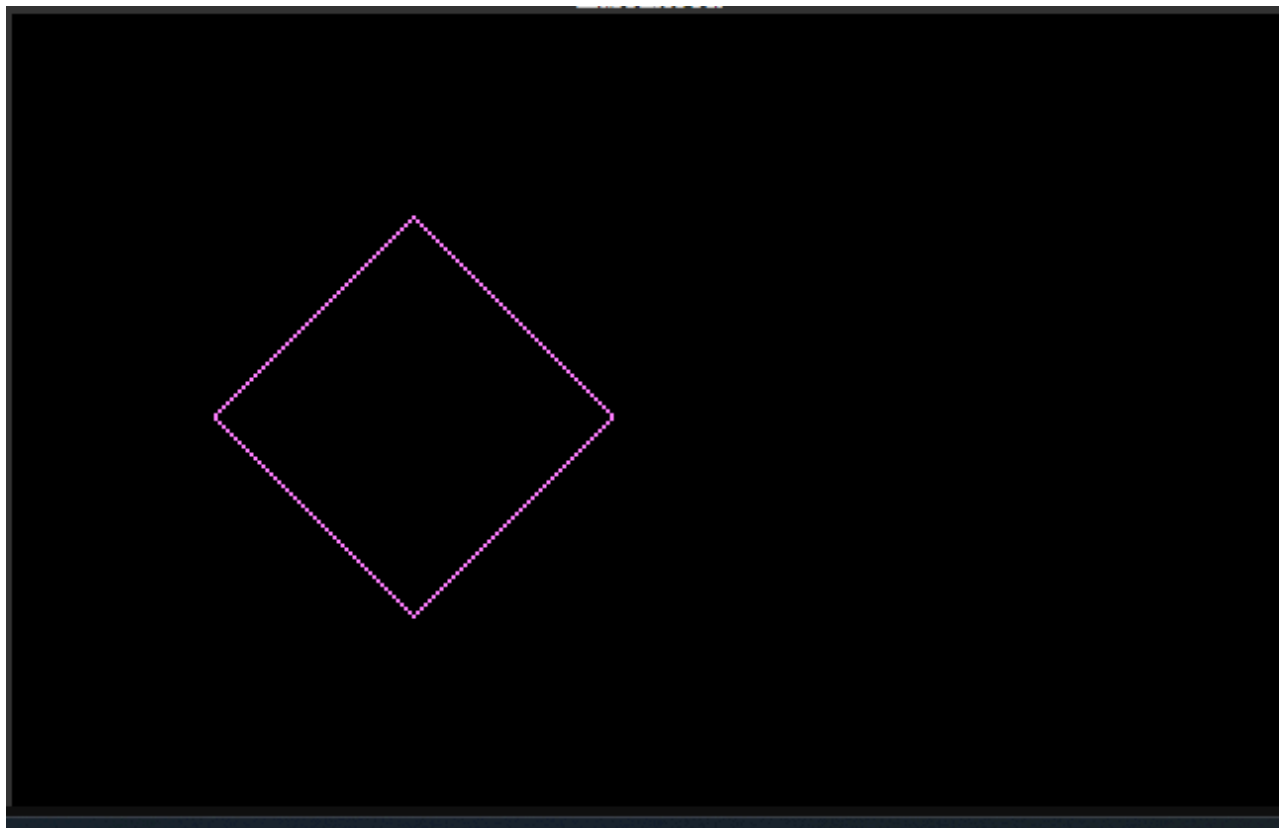
jbe u2

```
mov cx, y
mov dx, x-1
mov al, c
u3:
inc dx
mov ah, 0ch ; put pixel
int 10h
inc cx
cmp cx, y+w
jbe u3
```

```
mov cx, x+w+w
mov dx, y
mov al, c
u4:
inc dx
mov ah, 0ch ; put pixel
int 10h
dec cx
cmp dx, y+x
jbe u4
```

```
;wait for keypress
mov ah,00
int 16h
```

```
mov ax, 0x4c00
int 21h
```



Q3

[org 0x0100]

jmp code

w equ 30 ; width offset

x equ 30 ; starting x coordinate of line

y equ 30; starting y coordinate of line

c equ 60 ; color

x1 equ 50 ; starting x coordinate of line

y1 equ 100 ; starting y coordinate of line

code:

mov ah, 0

mov al, 13h

int 10h

; draw diagonal 11:

; N

mov cx, y

mov dx, x

mov al, c

u1:

inc dx

mov ah, 0ch ; put pixel

int 10h

cmp dx, x+w

jbe u1

mov cx, x

mov dx, y

mov al, c

u2:

inc dx

mov ah, 0ch ; put pixel

int 10h

inc cx

cmp dx, x+w

jbe u2

```
inc dx
u3:
dec dx
mov ah, 0ch ; put pixel
int 10h
cmp dx, x
jae u3
```

```
; E
mov cx, x+w+w
mov dx, y
mov al, c
```

```
u4:
inc dx
mov ah, 0ch ; put pixel
int 10h
cmp dx, x+w
jbe u4
```

```
mov cx, x+w+w
mov dx, y
mov al, c
u5:
mov ah, 0ch ; put pixel
int 10h
inc cx
cmp cx, x+w+w+w
jbe u5
```

```
mov cx, x+w+w
mov dx, y+15
mov al, c
u6:
mov ah, 0ch ; put pixel
int 10h
inc cx
cmp cx, x+w+w+w
jbe u6
```

```
mov cx, x+w+w
mov dx, y+w+1
```

```
mov al, c
u7:
mov ah, 0ch ; put pixel
int 10h
inc cx
cmp cx, x+w+w+w
jbe u7
```

```
; X
mov cx, x+w+w+w+w
mov dx, y
mov al, c
```

```
u8:
inc dx
mov ah, 0ch ; put pixel
int 10h
inc cx
cmp dx, x+w
jbe u8
```

```
mov cx, x+w+w+w+w+w
mov dx, y
mov al, c
u9:
inc dx
mov ah, 0ch ; put pixel
int 10h
dec cx
cmp dx, x+w
jbe u9
```

```
; T
mov cx, x+w+w+w+w+w+w
mov dx, y
mov al, c
```

```
u10:
mov ah, 0ch ; put pixel
int 10h
inc cx
cmp cx, x+w+w+w+w+w+w+w
jbe u10
```



```
mov cx, x+w+w+w+w+w+w+15
```

```
mov dx, y
```

```
mov al, c
```

```
u11:
```

```
inc dx
```

```
mov ah, 0ch ; put pixel
```

```
int 10h
```

```
cmp dx, y+w
```

```
jbe u11
```

```
mov ah,00
```

```
int 16h
```

```
mov ax, 0x4c00
```

```
int 21h
```



N E X T